Saeid Rahmani

Bioinformatics · Full-Stack Development · Machine Learning

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Summary

Computer engineer with an M.Sc. in Artificial Intelligence from Sharif University of Technology and experience in cancer genomics, unsupervised deep learning, and high-performance computing. Experienced in bulk and single-cell DNA/RNA analysis, clinical sequencing pipelines, research software, and highly available distributed systems. Research interests: multimodal learning from longitudinal clinical and molecular data to model cancer progression and treatment response.

Education

M.Sc. in Computer Engineering (Artificial Intelligence) — *Tehran, Iran*

Sharif University of Technology — *2017 - 2020*

Thesis: Unsupervised neural spike sorting by deep learning methods, advisor: Dr. Ali Sharifi-Zarchi

B.Sc. in Computer Engineering — *Tehran, Iran*

Kharazmi University — *2012 - 2016*

Advisor: Dr. Azadeh Mansouri

Skills

Machine Learning	PyTorch, TensorFlow, Keras, Scikit-learn; unsupervised deep learning; biomedical signal analysis
Bioinformatics	Bulk and single-cell DNA/RNA analysis, TCGA, HLA typing; Nextflow, Cromwell; Python, R
Computing	CUDA, MPI, OpenMP, Linux, Docker, Git; distributed systems; Java (Spring), React, C/C++
Languages	Farsi (native), English (fluent)

Work Experience

High-Performance Computing (HPC) Center — *Institute for Research in Fundamental Sciences (IPM), Tehran, Iran*

Full Stack Developer — *2022 - Present*

- Design and develop highly available distributed systems for a research computing platform (Java Spring, React, Docker).
- Manage HPC systems, computational job queues, and Linux system security.

Pardis Gene Genetic Lab — *Tehran, Iran*

Bioinformatician — *2022 - 2025*

- Developed HLA typing tools and bioinformatics pipelines (Nextflow, Cromwell) for clinical sequencing.
- Developed R/Python tools for clinical sequencing data analysis and biomarker research.

Selected Publications

InterOpt: Improved gene expression quantification in qPCR experiments using weighted aggregation of reference genes — *iScience, 2023*

A comprehensive in silico analysis and experimental validation of miRNAs capable of discriminating between lung adenocarcinoma and squamous cell carcinoma — *Frontiers in Genetics, 2024*

CLEC19A overexpression inhibits tumor cell proliferation/migration and promotes apoptosis... — *BMC Cancer, 2024*

Honors & Awards

1st — *Tehran, Iran*

The First National fNIRS Data Analysis Competition — *2021*

Best Paper Award — *Tehran, Iran*

Artificial Intelligence in Medical Imaging Conference — *2019*

2nd — *Iran*

National Master Entrance Exam (Iran) — *2017*

References

***Dr. Ali Sharifi-Zarchi*, Assistant Professor** — *asharifi@stanford.edu*

Sharif University of Technology

***Dr. Azadeh Mansouri*, Assistant Professor** — *a_mansouri@khu.ac.ir*

Kharazmi University

***Dr. Dara Rahmati*, Assistant Professor** — *d_rahmati@sbu.ac.ir*

Shahid Beheshti University